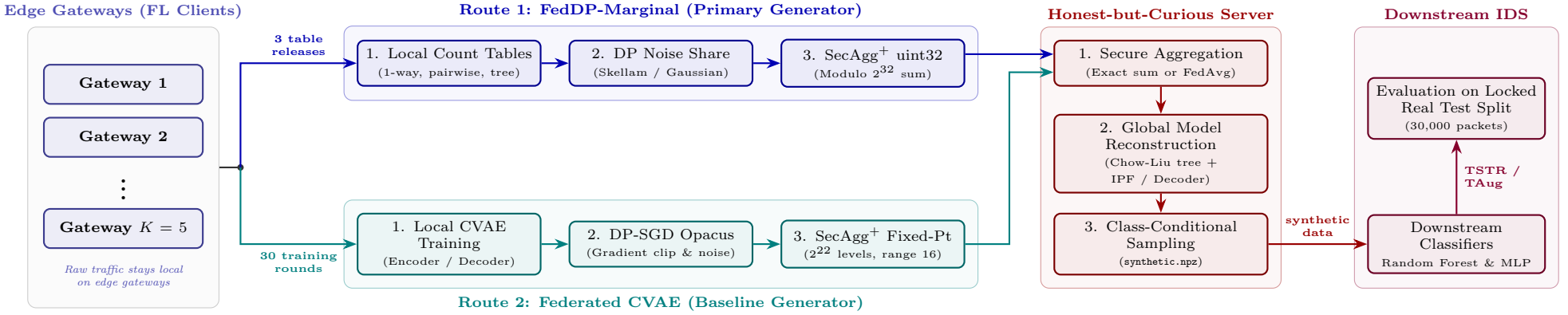

Edge Gateways (FL Clients)
Gateway 1
Gateway 2
Gateway $K = 5$
Raw traffic stays local on edge gateways
3 table releases
30 training rounds
Route 1: FedDP-Marginal (Primary Generator)
1. Local Count Tables
(1-way, pairwise, tree)
2. DP Noise Share
(Skellam / Gaussian)
3. SecAgg+ uint32
(Modulo 2^{32} sum)
1. Local CVAE Training
(Encoder / Decoder)
2. DP-SGD Opacus
(Gradient clip & noise)
3. SecAgg+ Fixed-Pt
(2^{22} levels, range 16)
Route 2: Federated CVAE (Baseline Generator)
Honest-but-Curious Server
1. Secure Aggregation
(Exact sum or FedAvg)
2. Global Model Reconstruction
(Chow-Liu tree + IPF / Decoder)
3. Class-Conditional Sampling
(synthetic.npz)
synthetic data
Downstream IDS
Evaluation on Locked Real Test Split
(30,000 packets)
TSTR / TAug
Downstream Classifiers
Random Forest & MLP